# EXPLORATORY DATA ANALYSIS PROJECTS

01 App Insights Unlocked
Google Play Store Data Analytics

02 Hotel Harmony
Data Insights for Optimized Hotel Operations

03 Pulse of Prevention
Analyzing Heart Health for Better Outcomes

**Prepared by: Raja Kumar**

Note: This document provides a complete project structure and analysis plan for all three projects listed in the supplied checklist. The checklist identifies the Google Play Store, hotel booking, and heart health topics, but it does not include the underlying datasets or detailed resource files. Numerical results must therefore be generated after the corresponding datasets are inserted and the code is executed.

# EDA01 - App Insights Unlocked: Google Play Store Data Analytics Using Python

## 1. Executive Summary

This project explores the Google Play Store ecosystem by examining app categories, ratings, reviews, installs, pricing, size, and user sentiment. The analysis combines an app catalogue containing 10,841 records and 13 fields with a user-review dataset containing 64,295 records and 5 fields. The aim is to understand what app performance looks like in the available data and turn those observations into practical questions for product, marketing, and app-development teams. The report distinguishes descriptive results from assumptions; it does not claim that a relationship automatically proves causation.

## 2. Project Background and Problem Statement

App stores contain many products competing for visibility and user attention. Ratings, review volume, installs, price, app size, category, and update activity can all provide useful signals, but these measures need to be examined together. The project addresses the need for a structured view of app performance and customer feedback rather than relying on a single metric.

## 3. Objectives

- Measure overall app ratings and review activity.
- Compare app categories and identify categories with strong install or rating patterns.
- Examine differences between free and paid apps.
- Explore relationships between app size, price, reviews, installs, and ratings.
- Review update dates and identify patterns in app maintenance.
- Use review sentiment fields and text to understand the tone of customer feedback.
- Present findings in a clear format that supports evidence-based product decisions.

## 4. Dataset Overview

| Dataset | Rows | Columns | Main contents |
|---|---|---|---|
| **Google Play Store apps** | 10,841 | 13 | App name, category, rating, reviews, size, installs, type, price, content rating, genre and update information |
| **Google Play Store user reviews** | 64,295 | 5 | App, translated review text, sentiment, polarity and subjectivity |

The source assignment identifies the Google Play Store dataset hosted on Kaggle. The analysis is limited to the fields present in the supplied CSV files.

# 5. Tools and Technologies

- Python for analysis and reproducible workflows.
- Pandas and NumPy for cleaning, transformation, aggregation, and calculations.
- Matplotlib and Seaborn for visual analysis.
- Jupyter Notebook or Google Colab for executing and documenting the work.

# 6. Methodology

## Step 1 — Inspect the data

Review row and column counts, data types, sample records, unique values, and missing-value patterns.

## Step 2 — Clean and standardize

Handle missing values carefully, remove exact duplicates where justified, standardize category and text values, and convert installs, prices, ratings, and dates into analysis-ready formats.

## Step 3 — Create useful metrics

Calculate average rating, review volume, install measures, price summaries, and category-level comparisons.

## Step 4 — Explore relationships

Compare ratings and reviews across app types, categories, size groups, and install levels. Use charts to make differences easier to interpret.

## Step 5 — Analyse customer feedback

Summarize sentiment categories and examine polarity/subjectivity where available. Review text-based results with care because automated labels may not capture context or sarcasm.

## Step 6 — Communicate results

Present key measures, visuals, limitations, and practical recommendations in a report or dashboard.

# 7. Initial Data Profile

The supplied app file contains 10,841 rows. In the raw file, the Rating column has 9,367 numeric values and an average of 4.19 where ratings are available. The Reviews column has 10,840 values that can be interpreted numerically after conversion. These are initial data-profile statistics; additional cleaning is required before treating them as final analytical results.
The five most represented categories by record count in the supplied file are: FAMILY (1,972); GAME (1,144); TOOLS (843); MEDICAL (463); BUSINESS (460).
Review sentiment labels in the supplied review file are distributed as follows: Positive: 23,998; Negative: 8,271; Neutral: 5,163.

## 8. Analysis Questions Covered

- Basic analysis: average rating, category distribution, app types, review counts, installs, and content ratings.
- Intermediate analysis: free versus paid apps, app size and performance, price and review activity, category comparisons, and update trends.
- Advanced analysis: install bands, category-level comparisons, and sentiment analysis using the user-review dataset.

## 9. Conclusion

This project provides a structured approach to understanding Google Play Store app performance and user feedback. By combining app-level metrics with review sentiment, the analysis can help identify patterns worth investigating and support more informed decisions about app quality, positioning, and customer experience. The strongest final conclusions should be based on the cleaned data, validated calculations, and visuals produced in the accompanying Python workbook.

# EDA02 - Hotel Harmony: Data Insights for Optimized Hotel Operations

## 1. Executive Summary

This project is designed to examine hotel booking and operational data to understand booking behaviour, cancellations, guest preferences, room pricing, revenue-related indicators, and seasonal patterns. The intended dataset is the hotel booking records described in the assignment. The project applies exploratory data analysis and, where appropriate, regression techniques to support operational decisions. The hotel CSV was not included with the materials available for preparing this report, so numerical findings are intentionally not invented. The report documents the project scope and the analysis process to be completed once the dataset is run.

## 2. Project Background and Problem Statement

Hotels need to balance occupancy, revenue, guest experience, and the risk of cancellations. Booking records can help managers understand when demand occurs, which channels bring reservations, which guest groups make special requests, and how room rates change across seasons. The challenge is to convert operational records into clear evidence that can guide planning and service improvements.

# 3. Project Objectives

- Understand booking volumes and patterns across hotel types and time periods.
- Measure cancellation and reservation-status patterns.
- Explore guest characteristics, booking channels, market segments, and special requests.
- Examine average daily rate (ADR) and related pricing patterns.
- Identify seasonal changes in demand and booking behaviour.
- Use statistical summaries and regression where appropriate to explore relationships with cancellations or rates.
- Develop practical recommendations for planning, revenue management, and guest service.

# 4. Dataset Overview

The assignment describes a hotel booking dataset commonly represented by fields such as hotel, is_canceled, lead_time, arrival year/month, weekend and weekday nights, adults/children/babies, meal, country, market segment, distribution channel, repeat-guest status, previous bookings, booking changes, deposit type, agent/company, waiting-list status, customer type, ADR, parking requests, special requests, reservation status, and reservation-status date. The exact fields and row count should be confirmed from the supplied CSV before analysis.

# 5. Tools and Technologies

- Python for repeatable analysis.
- Pandas and NumPy for cleaning, transformation, and aggregation.
- Matplotlib and Seaborn for charts.
- Scikit-learn or statsmodels for regression, if appropriate after data checks.
- Jupyter Notebook or Google Colab for documentation and execution.

# 6. Methodology

## Step 1 — Load and inspect

Read the CSV, confirm its dimensions and column names, inspect data types, and review sample rows.

## Step 2 — Validate data quality

Check missing values, duplicate records, impossible values, and inconsistent categories. Do not remove outliers automatically; investigate them first.

## Step 3 — Prepare date and booking fields

Convert arrival year/month and reservation-status dates into consistent types. Create derived measures such as total nights and arrival date when the required columns exist.

## Step 4 — Analyze bookings and cancellations

Calculate cancellation rates overall and compare them by hotel type, market segment, distribution channel, lead time, customer type, and deposit type.

### Step 5 — Analyze demand and pricing

Compare ADR and booking volumes by month, hotel type, and customer segment. Review seasonal patterns and possible outliers.

### Step 6 — Analyze guest behaviour

Examine repeat guests, special requests, parking requirements, family composition, and reservation changes.

### Step 7 — Model and communicate

If the dataset supports it, fit a suitable regression model with a train/test split and report metrics and limitations. Summarize results in charts and operational recommendations.

## 7. Analysis Questions Covered

- Basic analysis: booking counts, cancellation counts, guest composition, hotel types, booking channels, and summary statistics.
- Intermediate analysis: cancellations by booking attributes, monthly/seasonal patterns, ADR comparisons, repeat guests, and special requests.
- Advanced analysis: multivariable relationships and regression-based exploration of cancellation or pricing patterns, subject to data suitability.

## 8. Conclusion

Hotel Harmony is an exploratory analytics project intended to turn booking records into actionable operational insight. A careful workflow—from data validation to booking, cancellation, pricing, and guest-behaviour analysis—can help management make better-informed decisions. The final report should be updated with actual charts, metrics, and model results after the hotel CSV is available and analysed.

# EDA03 - Pulse of Prevention: Analyzing Heart Health for Better Outcomes

## 1. Executive Summary

This project explores a heart-health dataset using Python to understand how selected demographic and clinical measurements vary across records with and without the heart-disease target. The supplied file contains 1,025 rows and 14 columns. The analysis includes data inspection, descriptive statistics, category counts, group comparisons, visualizations, correlation analysis, and a demonstration logistic-regression model.

The work is intended to demonstrate a structured data-analysis process and identify patterns worth further investigation. The findings are descriptive associations in this dataset; they do not establish that any variable causes heart disease and should not be used to make medical decisions.

# 2. Introduction

Heart disease is an important public-health concern, and data analysis can help researchers examine patient characteristics and clinical measurements in a structured way. This project uses a tabular dataset containing fields such as age, sex code, chest-pain category, resting blood pressure, cholesterol, maximum heart rate, exercise-induced angina, and a binary target label.
Python provides a practical workflow for loading, validating, summarizing, visualizing, and modeling this information. The project follows a progression from basic descriptive analysis to more advanced statistical exploration and classification.

# 3. Project Goal and Objectives

The main goal is to analyze heart-health data using Python to explore factors associated with the heart-disease target and describe patient profiles in a clear, evidence-based way.

- Inspect the dataset structure, columns, data types, missing values, and repeated rows.
- Summarize age, blood pressure, cholesterol, heart-rate measures, and categorical variables.
- Compare selected measurements and category distributions across target groups.
- Use visualizations and correlation analysis to explore patterns among variables.
- Build and evaluate a baseline logistic-regression classifier as a learning exercise.
- Communicate limitations and avoid presenting the analysis as a clinical diagnostic system.

# 4. Dataset Description

The provided CSV has 1,025 rows and 14 columns. It contains 0 missing cells. A check identified 723 exact duplicate rows. These repeated rows are important: they should be investigated before modeling, because repeated records can make model evaluation look more optimistic than it would be on independent data.

| Column | Description |
|---|---|
| **age** | Age in years |
| **sex** | Encoded sex category; the project data dictionary describes 1 as male and 0 as female |
| **cp** | Chest-pain category code |
| **trestbps** | Resting blood pressure |
| **chol** | Serum cholesterol measurement |
| **fbs** | Fasting blood sugar indicator as encoded in the dataset |
| **restecg** | Resting electrocardiographic category |
| **thalach** | Maximum heart rate achieved |
| **exang** | Exercise-induced angina indicator |
| **oldpeak** | ST-depression-related measurement |
| **slope** | Peak-exercise ST-segment slope category |
| **ca** | Number/category of major vessels as encoded |
| **thal** | Thal-related category code |
| **target** | Binary label: 1 indicates heart disease present; 0 indicates absent |

The codes for categorical fields represent categories and should not automatically be treated as continuous measurements. The interpretation of each category should follow the project's supplied data dictionary.

# 5. Tools and Technologies

- Python: core programming language for the analysis.
- Pandas: data loading, cleaning, grouping, and tabular summaries.
- NumPy: numerical operations.
- Matplotlib and Seaborn: charts and visual exploration.
- SciPy: statistical testing, including Welch's independent-samples t-test.
- Scikit-learn: train/test splitting, feature scaling, logistic regression, and model evaluation.

# 6. Methodology

## Step 1 — Load the data

The CSV is loaded into a Pandas DataFrame and its shape, first rows, data types, and descriptive statistics are reviewed.

## Step 2 — Check data quality

Missing values and exact duplicate rows are counted. Numeric ranges and unique category codes are inspected. Duplicate rows are not automatically deleted without review.

## Step 3 — Perform descriptive analysis

Means, medians, counts, and percentages are calculated for demographic, clinical, and categorical fields.

## Step 4 — Explore relationships

Cross-tabulations, grouped summaries, box plots, count plots, scatter plots, and a correlation heatmap are used to compare variables and target groups.

## Step 5 — Statistical comparison

A Welch t-test is used as an exploratory comparison of resting blood pressure between the two encoded sex groups. The p-value is interpreted cautiously and does not imply causation.

## Step 6 — Build a baseline model

A logistic-regression pipeline is trained using a stratified train/test split, with scaling applied in the pipeline. Accuracy, ROC-AUC, classification report, and confusion matrix are used for evaluation.

## Step 7 — Interpret and report

Results are described as patterns in this dataset, alongside limitations and suggestions for more reliable follow-up analysis.7. Exploratory Analysis and Results

## 7.1 Dataset quality

The initial quality check found 0 missing cells across the dataset. It also identified 723 exact duplicate rows out of 1,025 rows. This is a substantial number of repeated records, so the source and duplication pattern should be checked before making strong claims or relying on a machine-learning score. This report does not silently remove those rows.

## 7.2 Descriptive statistics

| Measure | Observed value |
|---|---|
| **Average age** | 54.43 years |
| **Average resting blood pressure** | 131.61 (dataset units) |
| **Average cholesterol** | 246.00 (dataset units) |
| **Maximum thalach** | 202 |
| **Records with fbs = 1** | 153 |
| **Records with exang = 1** | 33.66% |
| **Records with restecg = 2** | 15 |

## 7.3 Encoded sex distribution

| Sex code | Record count | Percentage |
|---|---|---|
| **0** | 312 | 30.44% |
| **1** | 713 | 69.56% |

The values above describe the distribution of the encoded categories in the supplied file. They should be interpreted according to the dataset's code definitions rather than as a broader population estimate.

## 7.4 Chest-pain category distribution

| cp category code | Record count |
|---|---|
| **0** | 497 |
| **1** | 167 |
| **2** | 284 |
| **3** | 77 |

## 7.5 Distribution of ca

| ca category code | Record count |
|---|---|
| **0** | 578 |
| **1** | 226 |
| **2** | 134 |
| **3** | 69 |
| **4** | 18 |

## 7.6 Target distribution

| Target | Meaning | Count | Percentage |
|---|---|---|---|
| **0** | Heart disease absent | 499 | 48.68% |
| **1** | Heart disease present | 526 | 51.32% |

## 7.7 Numeric measurements by target

| Target | Mean age | Mean BP | Mean cholesterol | Mean thalach | Mean oldpeak |
|---|---|---|---|---|---|
| **0** | 56.57 | 134.11 | 251.29 | 139.13 | 1.60 |
| **1** | 52.41 | 129.25 | 240.98 | 158.59 | 0.57 |

The grouped averages are useful for identifying differences worth exploring. They are not evidence that any individual measurement is sufficient to predict or diagnose heart disease.

## 7.8 Correlation with target

| Feature | Pearson correlation with target |
|---|---|
| **oldpeak** | -0.438 |
| **exang** | -0.438 |
| **cp** | 0.435 |
| **thalach** | 0.423 |
| **ca** | -0.382 |
| **slope** | 0.346 |
| **thal** | -0.338 |
| **sex** | -0.280 |
| **age** | -0.229 |
| **trestbps** | -0.139 |
| **restecg** | 0.134 |
| **chol** | -0.100 |
| **fbs** | -0.041 |

Because the target is binary and several feature codes are categorical, Pearson correlations should be treated as an initial screening view, not a definitive ranking of medical importance.

## 7.9 Target rate by thal category

| thal category | Percentage with target = 1 |
|---|---|
| **0** | 42.86% |
| **1** | 32.81% |
| **2** | 75.74% |
| **3** | 21.95% |

These are target-label proportions by thal category, not survival rates. The dataset does not provide a follow-up period, survival status, or time-to-event field.

# 8. Visualizations

The accompanying Python workbook creates visualizations to support interpretation, including count plots for encoded categories, box plots for numeric measures by target, a scatter plot for age and cholesterol, and a correlation heatmap. Charts should be generated by running the notebook, then inserted into the final submission if required by the project instructions.

- Count plots: compare the number of records across category codes.
- Box plots: compare the spread and median of measurements between groups.
- Scatter plots: examine possible relationships between two numerical variables.
- Heatmap: provide a compact view of pairwise numeric correlations.

# 9. Logistic Regression — Model Approach

A baseline logistic-regression model can be trained to classify the dataset's target label using the available feature columns. The recommended workflow uses a stratified 80/20 train/test split, a StandardScaler, and LogisticRegression in a scikit-learn pipeline. Performance is evaluated with accuracy, ROC-AUC, precision, recall, F1-score, and a confusion matrix.

A model score should not be treated as proof of clinical usefulness. The large number of exact duplicate rows creates a risk of overly optimistic evaluation if related copies appear in both training and test sets. A stronger next step is to investigate duplicates, establish whether they are repeated copies or distinct observations, and evaluate on a clean, independent dataset. Any clinical use would require rigorous validation and qualified medical oversight.

Model metrics are not included as fixed values in this report because they should be calculated after the data-quality decision and the model is run in the notebook. This avoids presenting an unverified or potentially misleading score.

# 10. Key Findings

- The supplied dataset contains 1,025 rows and 14 columns, with no missing cells detected in the initial check.
- The initial check found 723 exact duplicate rows, making data provenance and duplicate handling a major consideration.
- The dataset's mean age is 54.43 years, average resting blood pressure is 131.61, and average cholesterol is 246.00.
- 33.66% of records have exang = 1 in the supplied dataset.
- Comparisons across target groups can reveal patterns in this file, but those patterns may not generalize to other populations.
- The dataset cannot support a genuine survival-rate calculation because no time-to-event or survival outcome is provided.

# 11. Conclusion

This project demonstrates a complete introductory-to-advanced Python workflow for exploring heart-health data, from loading and validating the CSV to summarizing variables, comparing groups, visualizing relationships, and outlining a logistic-regression model. The dataset provides a basis for descriptive analysis, but the high number of exact duplicate rows must be addressed before drawing stronger conclusions. The most responsible outcome is a transparent analysis that explains its evidence and limitations and identifies what should be investigated next.